



Department of Computer Science
Faculty of Computing

Project Task 4

(*MAKING THE CONNECTIONS – LAN and WAN*)

Programme	Bachelor of Computer Science (<i>Data Engineering</i>)
Subject Code	SECR1213
Subject Name	Network Communications
Session-Sem	2024/2025-1
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Section	02
Group	CircuitCrew (CC)
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Date	18 / 12 / 2024

Table of Content

Floor Plan with wiring.....	3
List of connection , patch cords and switch ports.....	4
Cable type and length.....	7
Minutes meeting #1.....	8
Minutes Meeting #2.....	9

FLOOR PLAN WITH WIRING

a) GROUND FLOOR

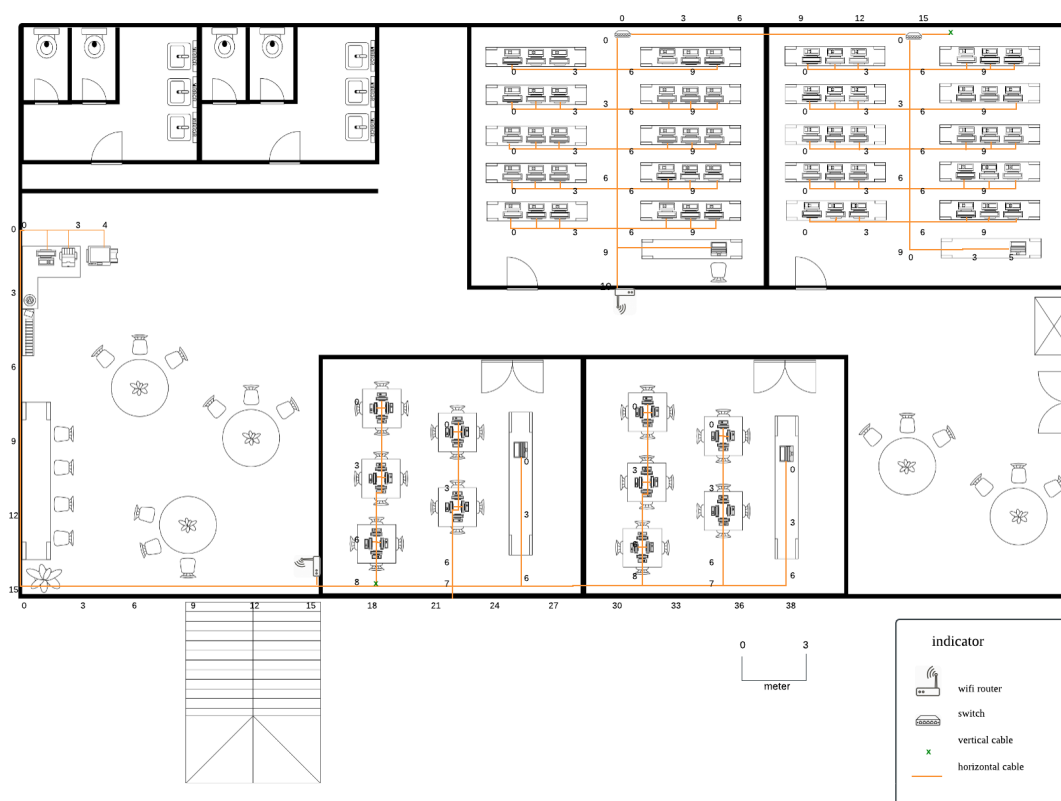


Figure 1: Ground floor plan with wiring

b) FIRST FLOOR

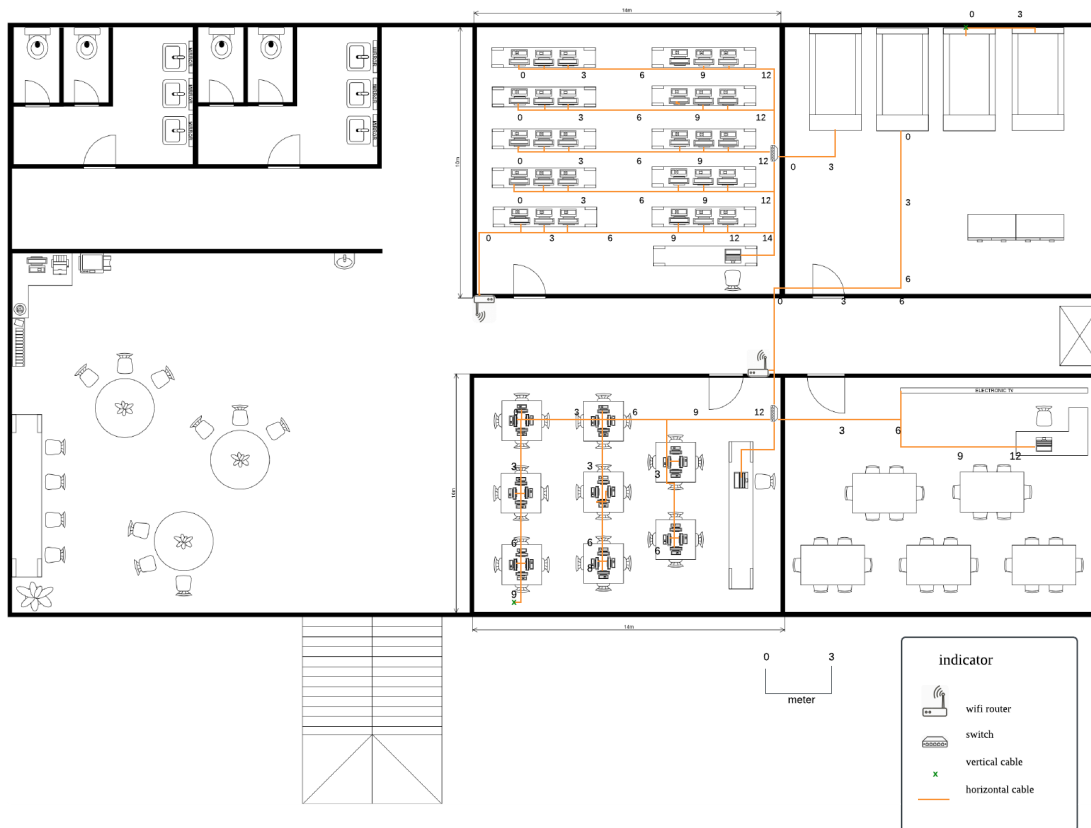


Figure 2: First floor plan with wiring

Figures 1 and Figure 2 shows two floor designs that represent the planned network connections between different rooms in the faculty building. The server room, where the router is located, will be the source of all connections. In order to establish a reliable connection for data transmission within the network, each room will be directly connected to the router through a backbone connection. The figure can also show how the wireless access points on the faculty building's ground and first floors are connected to one another inside the network.

CONNECTION, PATCH CORDS AND SWITCH PORTS

a) GROUND FLOOR

General-purpose Labs (2):

- Connections: 30 workstations each = 60 total.
- Switch Ports: $60 + (60 * 10\% \text{ spare}) = 66$ ports.
- Patch Cords: $60 + 10 \text{ spare} = 70$ cords.

Video Conferencing Room:

- Connections: Projector, workstation, access point = 3 total.
- Switch Ports: 4 (including spare).
- Patch Cords: 4 cords.

Student Lounge:

- Connections: 2 WiFi access points = 2 total.
- Switch Ports: 2 ports.
- Patch Cords: 2 cords.

b) FIRST FLOOR

Cisco Network Lab:

- Connections: 30 workstations + 5 networking devices = 35 total.
- Switch Ports: $35 + 10\% \text{ spare} = 39$ ports.
- Patch Cords: 40 cords.

Embedded Lab:

- Connections: 20 embedded systems + 5 IoT kits = 25 total.
- Switch Ports: $25 + 10\% \text{ spare} = 28$ ports.
- Patch Cords: 30 cords.

Hybrid Classroom:

- Connections: Projector, smartboard, 10 workstations = 12 total.
- Switch Ports: 12 + spare = 16 ports.
- Patch Cords: 16 cords.

Server Room:

- Connections: 5 servers + core network switches = 10 total.
- Switch Ports: 10 + spare = 16 ports.
- Patch Cords: 16 cords.

CABLE TYPE AND LENGTH

CABLE TYPE

1. Cat 6 Ethernet Cables:
 - ➔ Used for all workstation and device connections (provides up to 1 Gbps).
2. Fiber Optic Cables:
 - ➔ Used for backbone and vertical cabling between floors to support high-speed data transfer.
3. Cat 7 Ethernet Cables:
 - ➔ Used in Cisco Network Lab and IoT Lab for reduced interference and future-proofing.

CABLE LENGTH

1. General-purpose Labs:
 - ★ 10 meters per workstation = 600 meters (10m × 30 workstations × 2 labs).
2. Video Conferencing Room:
 - ★ 10 meters = 10 meters.
3. Student Lounge:
 - ★ 10 meters per access point = 20 meters.
4. Cisco Network Lab:
 - ★ 10 meters per device = 350 meters (including spares).
5. Embedded IoT Lab:
 - ★ 10 meters per device = 250 meters (including spares).
6. Hybrid Classroom:
 - ★ 10 meters per device = 120 meters.
7. Backbone Cabling:
 - ★ 50 meters per connection for vertical cabling = 100 meters (fiber optic).

Meeting Minutes #1

Date/Time		15 Dis 2024 / 5.00 pm	
Location		Foyer Dewan Resak	
Agenda		Task Distribution for Task 4	
Meeting Host		Damiya Aina binti Basir Abd Shammad	
Attendance			
Name		Time	Reason For Absence
Anis Safiyya		5.00 pm	-
Nurul Asyikin		5.00 pm	-
Damiya Aina		5.00 pm	-
Ain Nurnabila		5.00 pm	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1.	Read task 4	All members were assigned to read task 4 thoroughly	All members
2.	Floor plan	Asyikin and Ain in charge of drawing the cabling for floor plan	Nurul Asyikin and Ain Nurnabila
3.	Question 2	Damiya take the list of connection , patch cord and also switch ports	Damiya Aina
3.	Question 3	Anis take the question about the cable type and length	Anis Safiyya

Meeting Minutes #2

Date/Time		17 Dis 2024 / 9.30 pm	
Location		Whatsapp Chat	
Agenda		Progress Discussion	
Meeting Host		Damiya Aina binti Basir Abd Shammad	
Attendance			
Name		Time	Reason For Absence
Anis Safiyya		9.30 pm	-
Nurul Asyikin		9.30 pm	-
Damiya Aina		9.30 pm	-
Ain Nurnabila		9.30 pm	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1.	Sharing session on current progress	All of the members shared the current progress	All members
2.	Discuss the most suitable devices for the project	- All members share ideas and opinions of what devices that will be used for the project	All members
3.	Final edit in report	- Each member completing the additional information in report	All members
4.	Meeting ended	At 11.00	Damiya Aina